

Does College Pay Off?

Modeling the Return on Investment of U.S. Higher Education Using the College Scorecard

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1. Introduction

College tuition in the U.S. has climbed sharply over the last two decades, yet the financial payoff of a degree looks very different depending on where you go, what you study, where you live, and who you are. This project uses the U.S. Department of Education’s **College Scorecard** to take a data-driven look at that question.

We define **Return on Investment (ROI)** as the relationship between what a student pays (tuition, fees, and net price after financial aid) and what they earn after graduation.

Central Research Question: > *Which institutional characteristics and program features best predict whether a college degree delivers strong earnings relative to its cost, and how do these factors differ across institution types and fields of study?*

Hypotheses:

- **H1:** Net price meaningfully predicts median earnings 10 years after enrollment, even after controlling for selectivity.
 - **H2:** ROI varies significantly by institution type; for-profit schools deliver weaker earnings relative to cost.
 - **H3:** Field of study is a stronger earnings predictor than institutional characteristics.
 - **H4:** Higher graduation rates predict better earnings even after controlling for selectivity.
-

2. Setup & Libraries

```
library(MASS)           # load before tidyverse to avoid masking dplyr::select
library(tidyverse)
library(corrplot)
library(caret)
library(pROC)
library(factoextra)
library(scales)
library(knitr)
library(kableExtra)
```

3. Data Acquisition & Preprocessing

3.1 Load Data

Data downloaded from the U.S. Department of Education College Scorecard.

```
# Load institution-level data
# Download from: https://collegescorecard.ed.gov/data/
# File: Most Recent Institution-Level Data (CSV)
inst_raw <- read_csv(
  "data/Most-Recent-Cohorts-Institution.csv",
  na = c("", "NA", "PrivacySuppressed"),
  show_col_types = FALSE
)

# Load field-of-study data
# File: Most Recent Field of Study Data (CSV)
fos_raw <- read_csv(
  "data/Most-Recent-Cohorts-Field-of-Study.csv",
  na = c("", "NA", "PrivacySuppressed"),
  show_col_types = FALSE
)

cat("Institution-level rows:", nrow(inst_raw), "\n")
```

```
## Institution-level rows: 6322
```

```
cat("Field-of-study rows:  ", nrow(fos_raw),  "\n")
```

```
## Field-of-study rows:    227980
```

3.2 Filter & Select Key Variables

```
inst <- inst_raw |>
  # Keep only predominantly bachelor's-degree-granting institutions
  filter(PREDDEG == 3) |>
  # Select key variables
  select(
    UNITID, OPEID, INSTNM, STABBR, REGION,
    CONTROL, PREDDEG, UGDS,
    ADM_RATE, SAT_AVG,
    NPT4_PUB, NPT4_PRIV,
    C150_4,
    MD_EARN_WNE_P10,
    DEBT_MDN,
    starts_with("PCIP")
  ) |>
  # Coerce numeric columns
  mutate(across(c(ADM_RATE, SAT_AVG, NPT4_PUB, NPT4_PRIV,
                  C150_4, MD_EARN_WNE_P10, DEBT_MDN, UGDS,
                  starts_with("PCIP")), as.numeric)) |>
  # Drop rows with suppressed earnings outcome
  filter(!is.na(MD_EARN_WNE_P10))

cat("Institutions after filtering:", nrow(inst), "\n")
```

```
## Institutions after filtering: 1808
```

3.3 Feature Engineering

```
inst <- inst |>
  mutate(
    # Unified net price: use public rate for public schools, private for others
    net_price = case_when(
      CONTROL == 1 ~ NPT4_PUB,
      CONTROL %in% c(2, 3) ~ NPT4_PRIV,
      TRUE ~ NA_real_
    ),
    # ROI: earnings 10yr after entry divided by net price
    ROI = MD_EARN_WNE_P10 / net_price,
    # Binary ROI label for classification
    high_roi = ifelse(ROI > median(ROI, na.rm = TRUE), 1, 0),
    # STEM-heavy flag: sum of STEM-related PCIP field shares
    stem_share = rowSums(
      pick(PCIP11, PCIP14, PCIP15, PCIP26, PCIP27, PCIP40),
      na.rm = TRUE
    ),
    is_stem_heavy = as.integer(stem_share > 0.3),
    # Readable institution type label
    inst_type = case_when(
      CONTROL == 1 ~ "Public",
      CONTROL == 2 ~ "Private Nonprofit",
      CONTROL == 3 ~ "Private For-Profit",
      TRUE ~ NA_character_
    ),
    # Log-transform skewed variables for modeling
    log_ugds = log1p(UGDS),
    log_net_price = log1p(net_price),
    log_earnings = log1p(MD_EARN_WNE_P10)
  )

cat("Rows with valid ROI:", sum(!is.na(inst$ROI)), "\n")
```

```
## Rows with valid ROI: 1695
```

3.4 Missingness Analysis

```
miss_summary <- inst |>
  select(net_price, ADM_RATE, SAT_AVG, C150_4,
         MD_EARN_WNE_P10, DEBT_MDN, ROI) |>
  summarise(across(everything(), ~mean(is.na(.)) * 100)) |>
  pivot_longer(everything(), names_to = "variable", values_to = "pct_missing") |>
  arrange(desc(pct_missing))

miss_summary |>
  kable(digits = 1, col.names = c("Variable", "% Missing"),
        caption = "Missing Data Summary for Key Variables") |>
  kable_styling(bootstrap_options = c("striped", "hover"), full_width = FALSE)
```

Table 1: Missing Data Summary for Key Variables

Variable	% Missing
SAT_AVG	45.2
ADM_RATE	16.4
net_price	6.2
ROI	6.2
C150_4	5.8
DEBT_MDN	2.3
MD_EARN_WNE_P10	0.0

```
# Drop variables with > 40% missing
vars_to_drop <- miss_summary |> filter(pct_missing > 40) |> pull(variable)
cat("Variables dropped (>40% missing):", paste(vars_to_drop, collapse = ", "), "\n")
```

```
## Variables dropped (>40% missing): SAT_AVG
```

```
# Complete-case analysis on core modeling variables
model_df <- inst |>
  filter(!is.na(net_price), !is.na(ADM_RATE), !is.na(SAT_AVG),
         !is.na(C150_4), !is.na(MD_EARN_WNE_P10), !is.na(ROI)) |>
  mutate(REGION = factor(REGION), inst_type = factor(inst_type))

cat("Complete cases for modeling:", nrow(model_df), "\n")
```

```
## Complete cases for modeling: 990
```

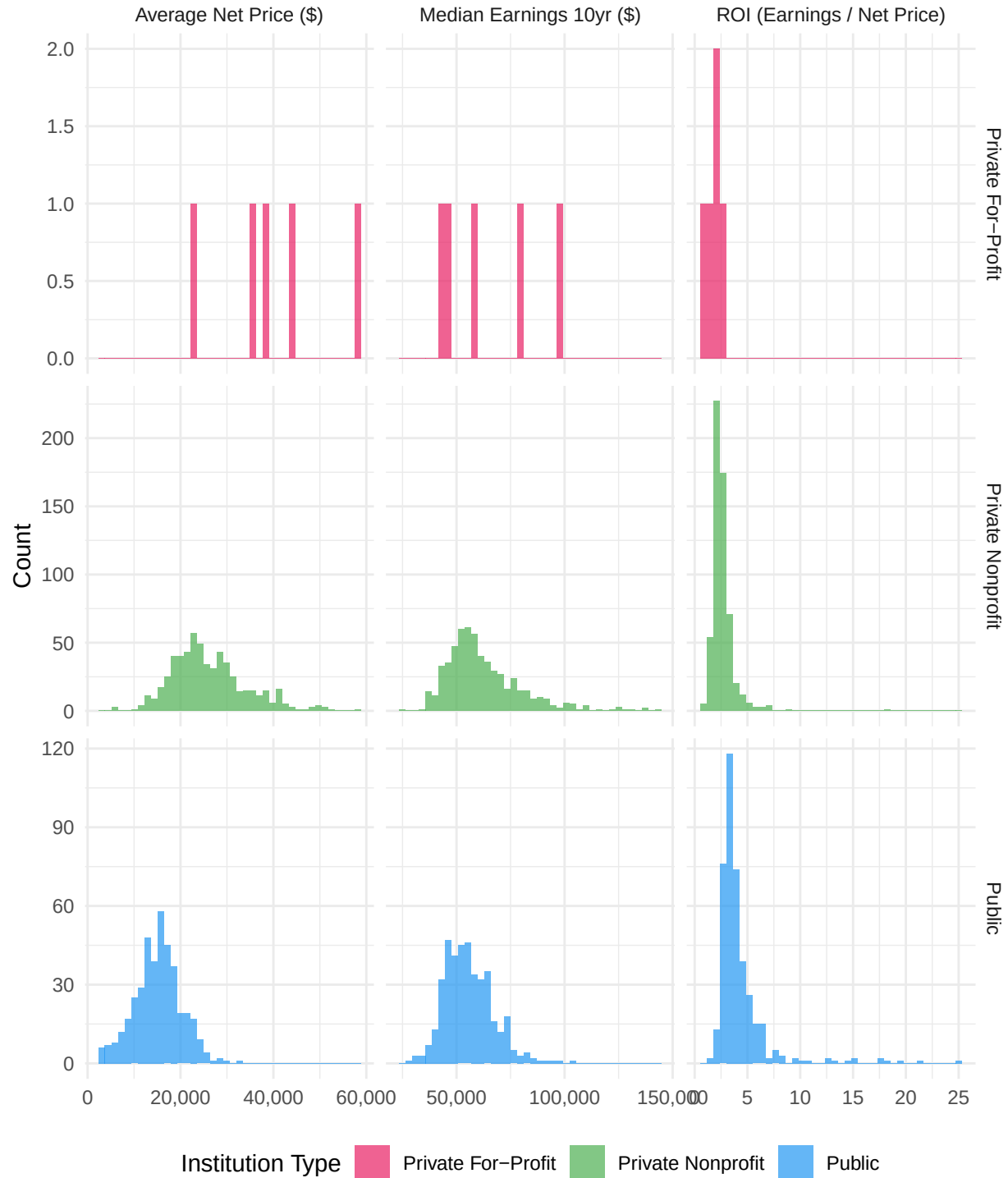
4. Exploratory Data Analysis

4.1 Distribution of Key Variables

```
dist_data <- model_df |>
  select(inst_type, net_price, MD_EARN_WNE_P10, ROI) |>
  pivot_longer(-inst_type, names_to = "variable", values_to = "value") |>
  mutate(variable = recode(variable,
    net_price      = "Average Net Price ($)",
    MD_EARN_WNE_P10 = "Median Earnings 10yr ($)",
    ROI            = "ROI (Earnings / Net Price)"
  ))

ggplot(dist_data, aes(x = value, fill = inst_type)) +
  geom_histogram(bins = 40, alpha = 0.7, position = "identity") +
  facet_grid(inst_type ~ variable, scales = "free") +
  scale_x_continuous(labels = label_comma()) +
  scale_fill_manual(values = c("Public" = "#2196F3",
    "Private Nonprofit" = "#4CAF50",
    "Private For-Profit" = "#E91E63")) +
  labs(title = "Distribution of Net Price, Earnings, and ROI by Institution Type",
    x = NULL, y = "Count", fill = "Institution Type") +
  theme_minimal(base_size = 11) +
  theme(legend.position = "bottom", strip.text = element_text(size = 9))
```

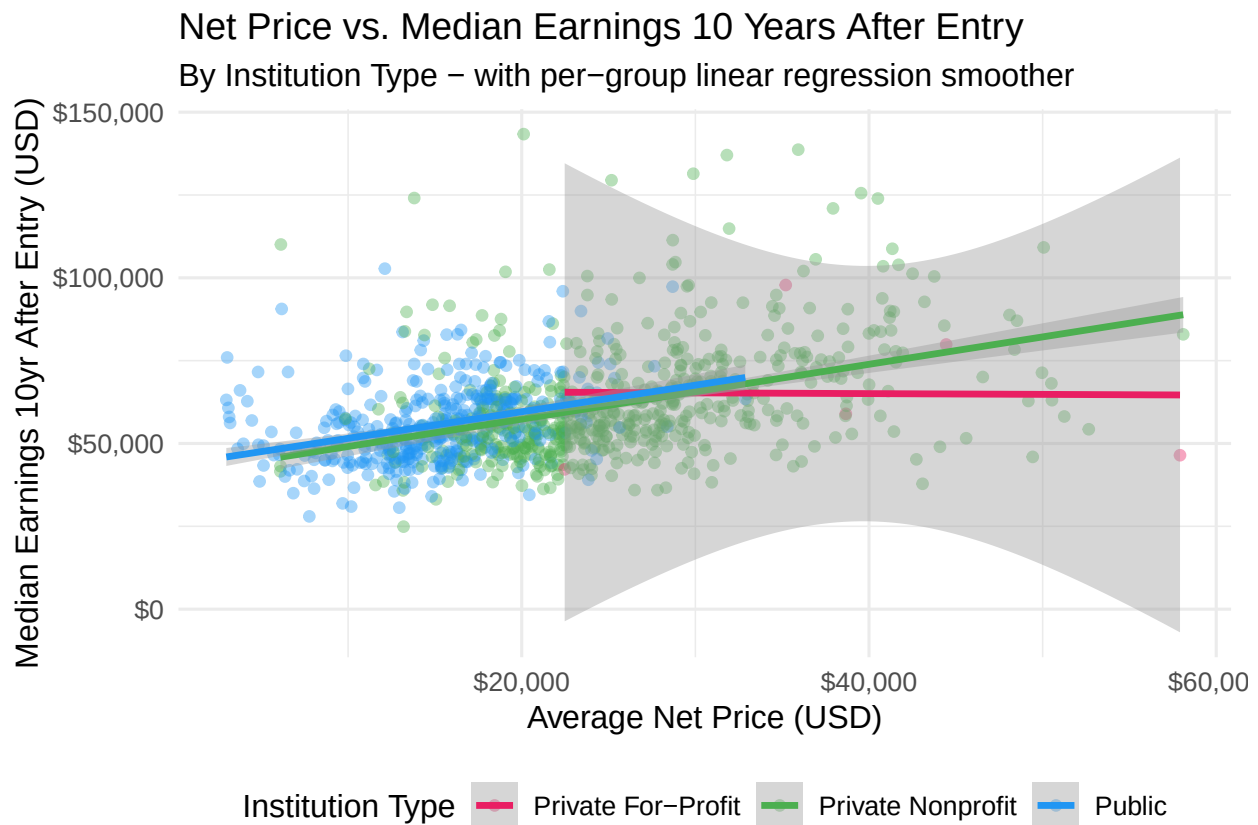
Distribution of Net Price, Earnings, and ROI by Institution Type



4.2 Net Price vs. Median Earnings

```
ggplot(model_df, aes(x = net_price, y = MD_EARN_WNE_P10, color = inst_type)) +
  geom_point(alpha = 0.4, size = 1.5) +
```

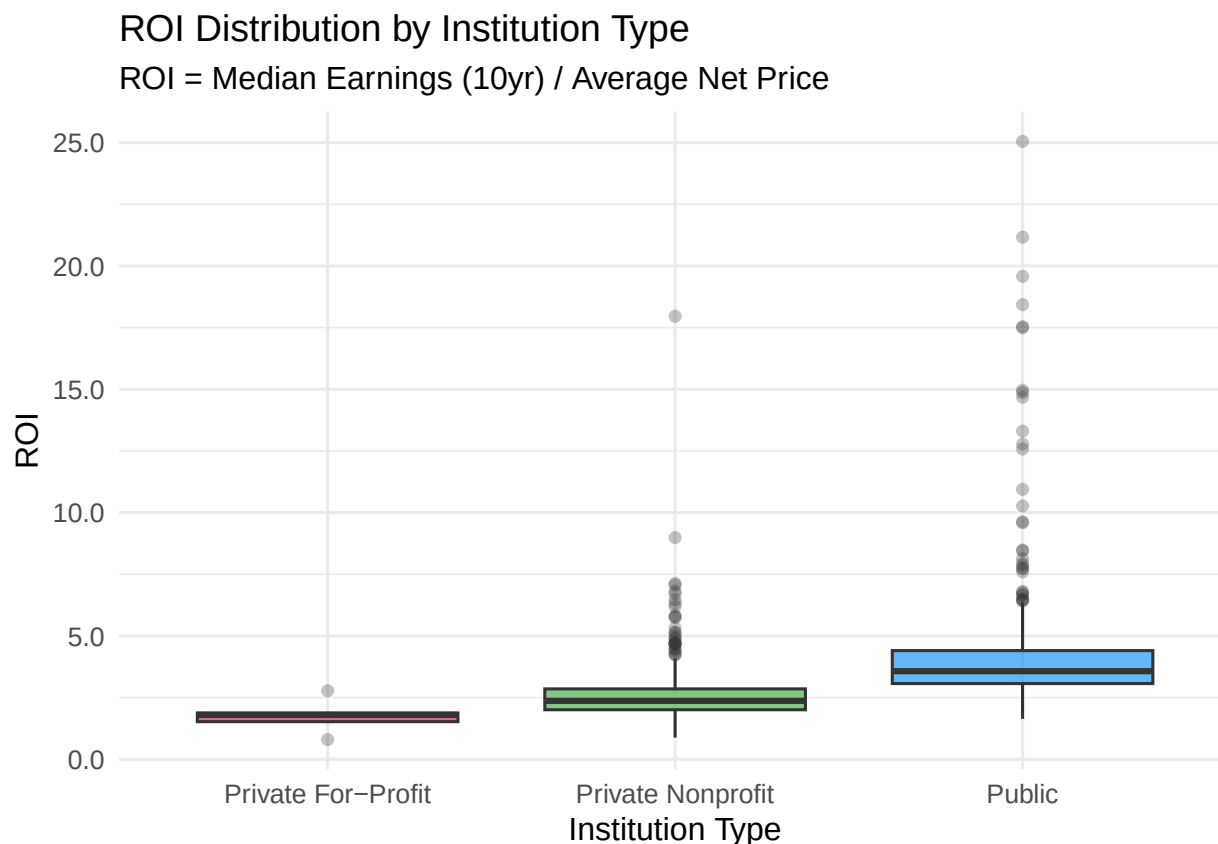
```
geom_smooth(method = "lm", se = TRUE, linewidth = 1.2) +
scale_x_continuous(labels = dollar_format()) +
scale_y_continuous(labels = dollar_format()) +
scale_color_manual(values = c("Public" = "#2196F3",
                              "Private Nonprofit" = "#4CAF50",
                              "Private For-Profit" = "#E91E63")) +
labs(title = "Net Price vs. Median Earnings 10 Years After Entry",
     subtitle = "By Institution Type - with per-group linear regression smoother",
     x = "Average Net Price (USD)", y = "Median Earnings 10yr After Entry (USD)",
     color = "Institution Type") +
theme_minimal(base_size = 12) +
theme(legend.position = "bottom")
```



4.3 ROI by Institution Type

```
ggplot(model_df, aes(x = inst_type, y = ROI, fill = inst_type)) +
geom_boxplot(alpha = 0.7, outlier.alpha = 0.3) +
scale_fill_manual(values = c("Public" = "#2196F3",
                              "Private Nonprofit" = "#4CAF50",
                              "Private For-Profit" = "#E91E63")) +
scale_y_continuous(labels = label_number(accuracy = 0.1)) +
labs(title = "ROI Distribution by Institution Type",
     subtitle = "ROI = Median Earnings (10yr) / Average Net Price",
     x = "Institution Type", y = "ROI", fill = NULL) +
theme_minimal(base_size = 12) +
```

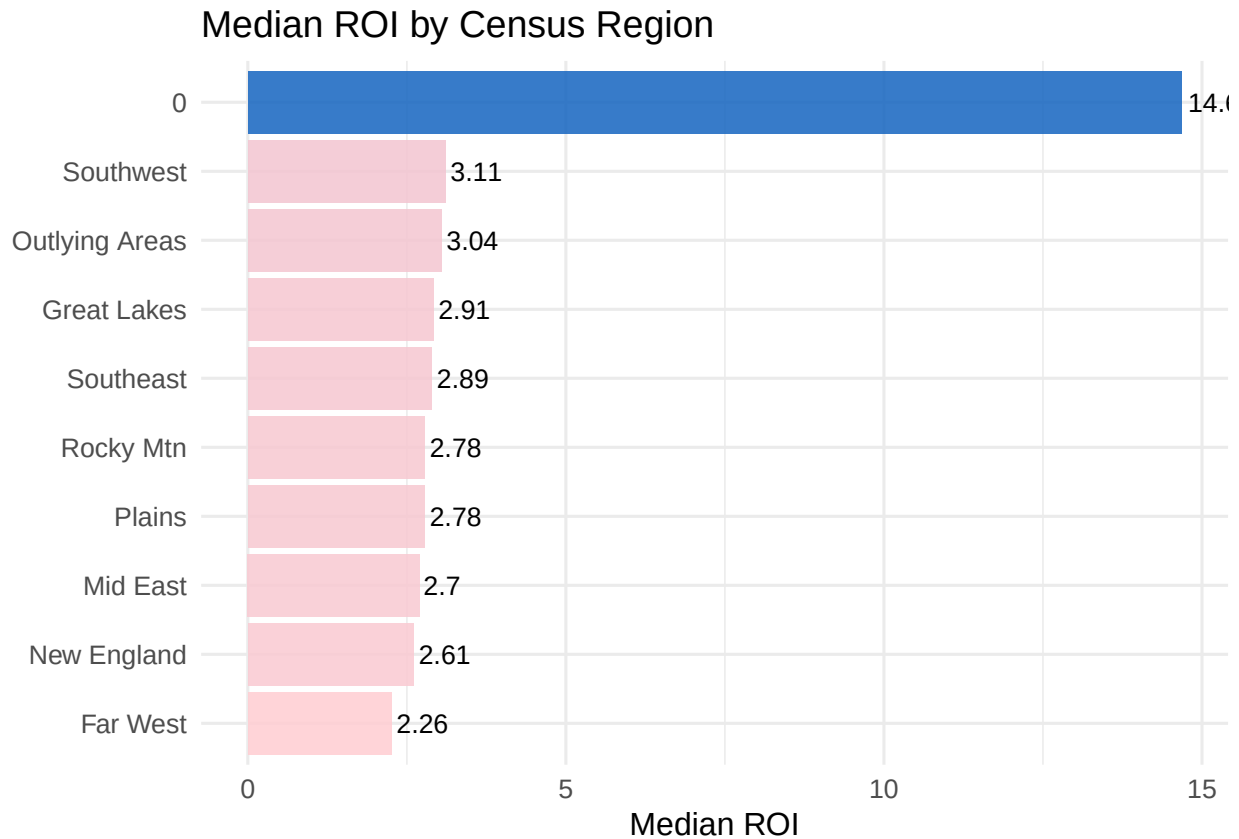
```
theme(legend.position = "none")
```



4.4 Geographic Analysis — Median ROI by Region

```
region_labels <- c(
  "1" = "New England", "2" = "Mid East", "3" = "Great Lakes",
  "4" = "Plains", "5" = "Southeast", "6" = "Southwest",
  "7" = "Rocky Mtn", "8" = "Far West", "9" = "Outlying Areas"
)

model_df |>
  mutate(region_name = recode(as.character(REGION), !!!region_labels)) |>
  group_by(region_name) |>
  summarise(median_roi = median(ROI, na.rm = TRUE),
            n = n(), .groups = "drop") |>
  arrange(desc(median_roi)) |>
  ggplot(aes(x = fct_reorder(region_name, median_roi), y = median_roi, fill = median_roi)) +
  geom_col(alpha = 0.85) +
  geom_text(aes(label = round(median_roi, 2)), hjust = -0.1, size = 3.5) +
  scale_fill_gradient(low = "#FFCDD2", high = "#1565C0") +
  coord_flip() +
  labs(title = "Median ROI by Census Region",
       x = NULL, y = "Median ROI", fill = "Median ROI") +
  theme_minimal(base_size = 12) +
  theme(legend.position = "none")
```



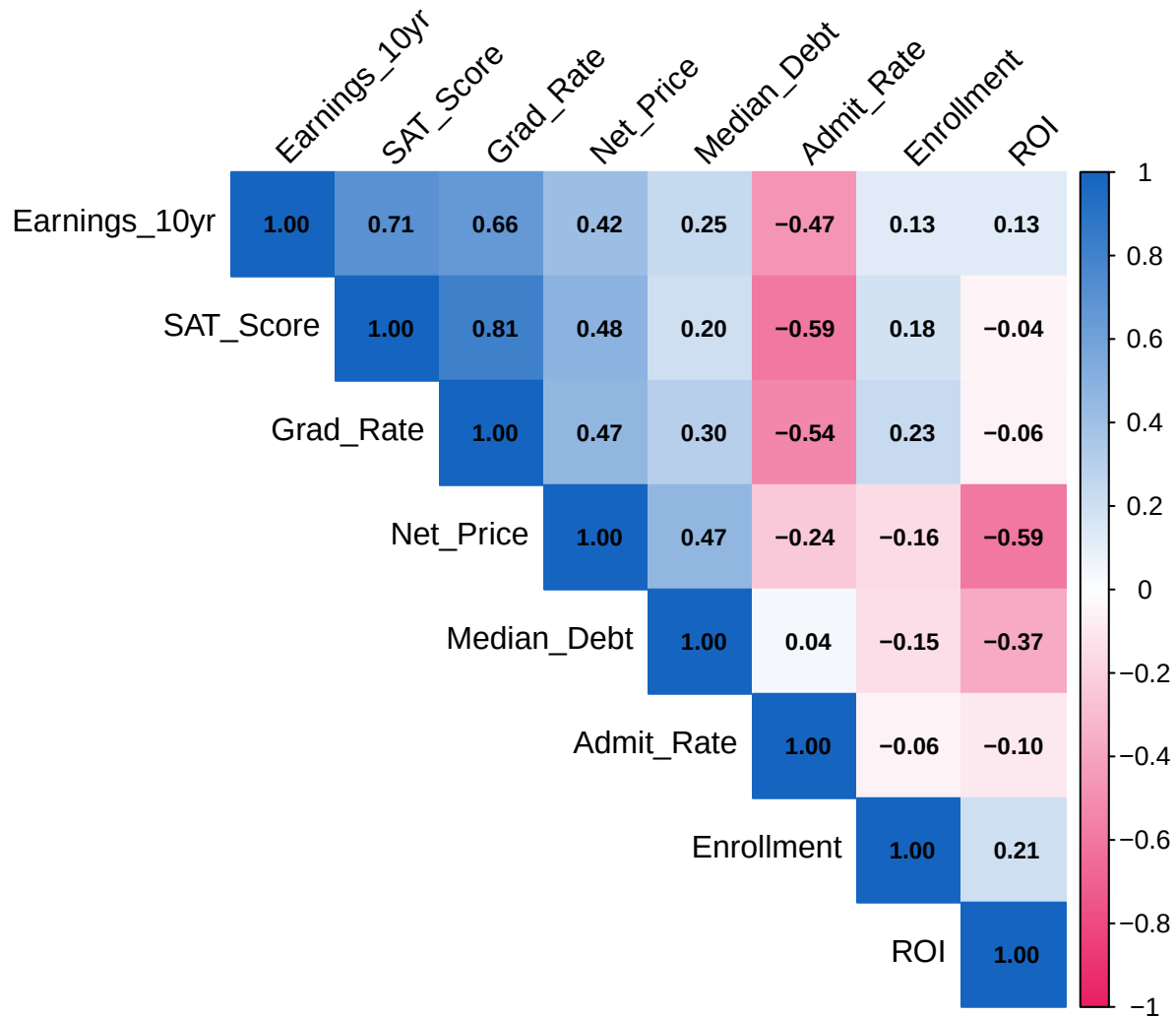
4.5 Correlation Heatmap

```
corr_vars <- model_df |>
  select(MD_EARN_WNE_P10, net_price, ADM_RATE, SAT_AVG,
         C150_4, UGDS, DEBT_MDN, ROI) |>
  rename(Earnings_10yr = MD_EARN_WNE_P10,
         Net_Price     = net_price,
         Admit_Rate    = ADM_RATE,
         SAT_Score     = SAT_AVG,
         Grad_Rate     = C150_4,
         Enrollment    = UGDS,
         Median_Debt   = DEBT_MDN)

corr_matrix <- cor(corr_vars, use = "pairwise.complete.obs")

corrplot(corr_matrix,
         method = "color", type = "upper", order = "hclust",
         tl.col = "black", tl.srt = 45,
         addCoef.col = "black", number.cex = 0.75,
         col = colorRampPalette(c("#E91E63", "white", "#1565C0"))(200),
         title = "\nCorrelation Matrix of Key Variables")
```

Correlation Matrix of Key Variables



4.6 Earnings by Field of Study

```
# PCIP field labels (CIP codes)
pcip_labels <- tribble(
  ~code, ~field,
  "PCIP01", "Agriculture",
  "PCIP03", "Natural Resources",
  "PCIP04", "Architecture",
  "PCIP05", "Area Studies",
  "PCIP09", "Communication",
```

```

"PCIP10", "Comm. Technologies",
"PCIP11", "Computer Science",
"PCIP12", "Personal Services",
"PCIP13", "Education",
"PCIP14", "Engineering",
"PCIP15", "Engineering Tech",
"PCIP16", "Foreign Languages",
"PCIP19", "Family Sciences",
"PCIP22", "Legal Studies",
"PCIP23", "English",
"PCIP24", "Liberal Arts",
"PCIP25", "Library Science",
"PCIP26", "Biology",
"PCIP27", "Mathematics",
"PCIP29", "Military Studies",
"PCIP30", "Interdisciplinary",
"PCIP31", "Leisure Studies",
"PCIP38", "Philosophy",
"PCIP39", "Theology",
"PCIP40", "Physical Sciences",
"PCIP41", "Science Tech",
"PCIP42", "Psychology",
"PCIP43", "Criminal Justice",
"PCIP44", "Public Admin",
"PCIP45", "Social Sciences",
"PCIP46", "Construction",
"PCIP47", "Mechanic/Repair",
"PCIP48", "Precision Production",
"PCIP49", "Transportation",
"PCIP50", "Visual/Performing Arts",
"PCIP51", "Health Professions",
"PCIP52", "Business",
"PCIP54", "History"
)

# Weighted average earnings by dominant field
field_earnings <- model_df |>
  select(MD_EARN_WNE_P10, starts_with("PCIP")) |>
  pivot_longer(-MD_EARN_WNE_P10, names_to = "code", values_to = "share") |>
  filter(!is.na(share), share > 0) |>
  left_join(pcip_labels, by = "code") |>
  group_by(field) |>
  summarise(
    weighted_earnings = weighted.mean(MD_EARN_WNE_P10, share, na.rm = TRUE),
    n = n(),
    .groups = "drop"
  ) |>
  filter(!is.na(field), n >= 10) |>
  arrange(desc(weighted_earnings))

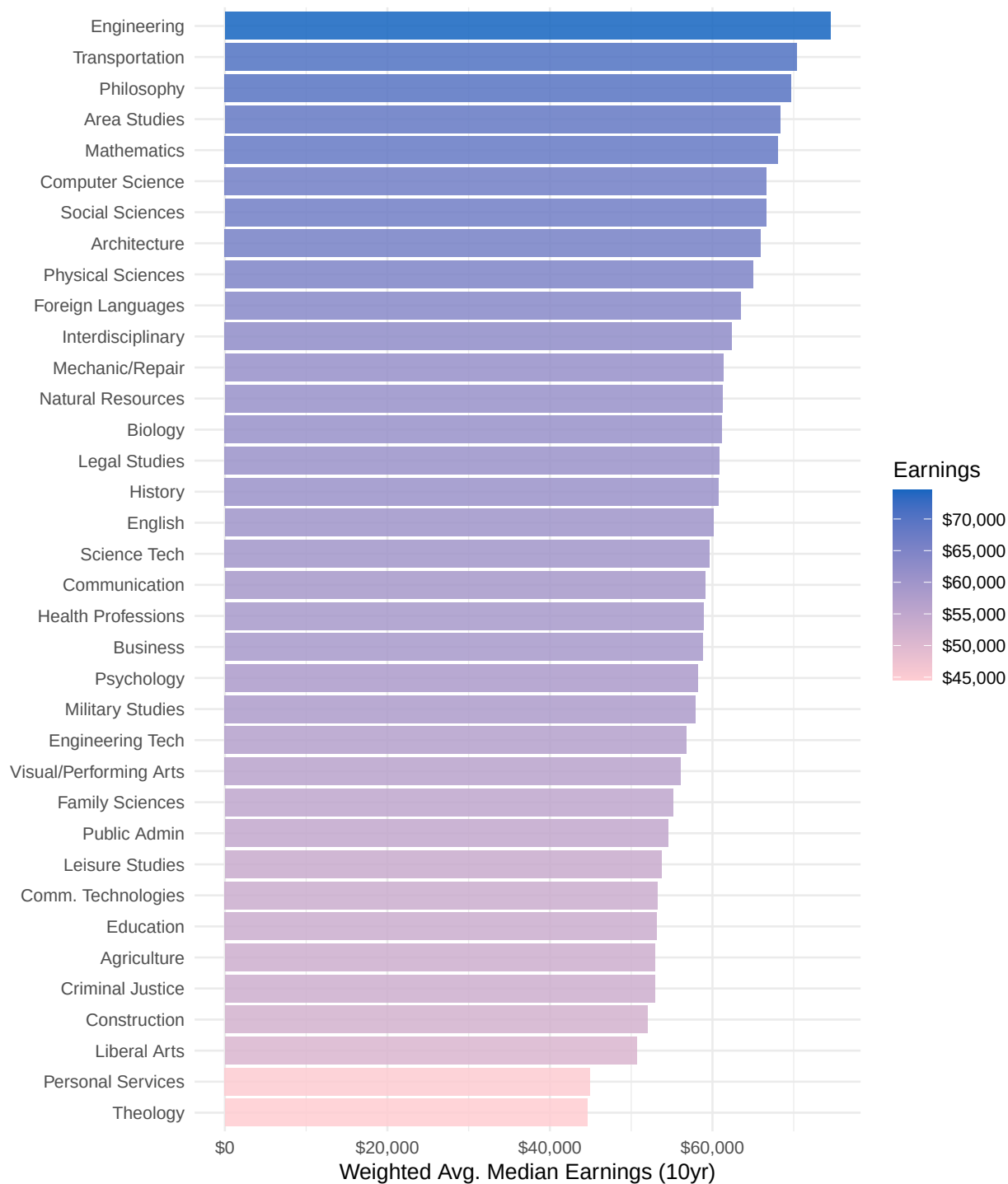
ggplot(field_earnings, aes(x = fct_reorder(field, weighted_earnings),
                           y = weighted_earnings, fill = weighted_earnings)) +
  geom_col(alpha = 0.85) +

```

```
scale_y_continuous(labels = dollar_format()) +  
scale_fill_gradient(low = "#FFCDD2", high = "#1565C0", labels = dollar_format()) +  
coord_flip() +  
labs(title = "Weighted Average Earnings by Field of Study",  
      subtitle = "Weighted by fraction of degrees awarded in each field",  
      x = NULL, y = "Weighted Avg. Median Earnings (10yr)", fill = "Earnings") +  
theme_minimal(base_size = 10)
```

Weighted Average Earnings by Field of Study

Weighted by fraction of degrees awarded in each field



5. Hypothesis Testing

5.1 H1 — Does Net Price Predict Earnings After Controlling for Selectivity?

```
# Base model: selectivity only
h1_base <- lm(MD_EARN_WNE_P10 ~ ADM_RATE + SAT_AVG, data = model_df)

# Full model: add net price
h1_full <- lm(MD_EARN_WNE_P10 ~ ADM_RATE + SAT_AVG + net_price, data = model_df)

summary(h1_full)
```

```
##
## Call:
## lm(formula = MD_EARN_WNE_P10 ~ ADM_RATE + SAT_AVG + net_price,
##     data = model_df)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -38959  -6007   -808    4984   80204
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -1.708e+04  4.499e+03  -3.796 0.000156 ***
## ADM_RATE     -5.400e+03  1.893e+03  -2.853 0.004423 **
## SAT_AVG       6.447e+01  3.275e+00  19.688 < 2e-16 ***
## net_price     1.840e-01  4.449e-02   4.135 3.85e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 10940 on 986 degrees of freedom
## Multiple R-squared:  0.5099, Adjusted R-squared:  0.5084
## F-statistic: 341.9 on 3 and 986 DF,  p-value: < 2.2e-16
```

```
# Partial F-test: does adding net price significantly improve fit?
anova(h1_base, h1_full)
```

```
## Analysis of Variance Table
##
## Model 1: MD_EARN_WNE_P10 ~ ADM_RATE + SAT_AVG
## Model 2: MD_EARN_WNE_P10 ~ ADM_RATE + SAT_AVG + net_price
##   Res.Df    RSS Df Sum of Sq    F    Pr(>F)
## 1      987 1.2009e+11
## 2      986 1.1805e+11  1 2047132601 17.099 3.85e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Interpretation: If the `net_price` coefficient is statistically significant ($p < 0.05$) and the ANOVA partial F-test shows a significant improvement, H1 is supported — net price predicts earnings even after accounting for selectivity.

5.2 H2 — Does ROI Differ Significantly by Institution Type?

```
# One-way ANOVA
h2_aov <- aov(ROI ~ inst_type, data = model_df)
```

```
summary(h2_aov)

##              Df Sum Sq Mean Sq F value Pr(>F)
## inst_type      2     706    353.1    98.4 <2e-16 ***
## Residuals    987    3541     3.6
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

# Post-hoc pairwise comparisons
as.data.frame(TukeyHSD(h2_aov)$inst_type) |>
  round(4) |>
  kable(caption = "Tukey HSD Post-hoc Comparisons for ROI by Institution Type") |>
  kable_styling(bootstrap_options = c("striped", "hover"), full_width = FALSE)
```

Table 2: Tukey HSD Post-hoc Comparisons for ROI by Institution Type

	diff	lwr	upr	p adj
Private Nonprofit-Private For-Profit	0.8130	-1.1839	2.8099	0.6052
Public-Private For-Profit	2.5205	0.5198	4.5211	0.0089
Public-Private Nonprofit	1.7075	1.4194	1.9955	0.0000

Interpretation: A significant ANOVA F-test ($p < 0.05$) supports H2. Tukey post-hoc tests identify which specific pairs of institution types differ significantly in ROI.

5.3 H3 — Field of Study vs. Institution Characteristics

```
# Model 1: only institutional characteristics
h3_inst <- lm(MD_EARN_WNE_P10 ~ ADM_RATE + SAT_AVG + net_price +
  inst_type + UGDS + REGION,
  data = model_df)

# Model 2: add STEM indicator (proxy for field of study)
h3_field <- lm(MD_EARN_WNE_P10 ~ ADM_RATE + SAT_AVG + net_price +
  inst_type + UGDS + REGION + stem_share + is_stem_heavy,
  data = model_df)

# Compare R²
cat("Institutional model R²: ", round(summary(h3_inst)$r.squared, 4), "\n")

## Institutional model R²: 0.5709
cat("Institutional + Field model R²: ", round(summary(h3_field)$r.squared, 4), "\n")

## Institutional + Field model R²: 0.6337

# Partial F-test
anova(h3_inst, h3_field)

## Analysis of Variance Table
##
## Model 1: MD_EARN_WNE_P10 ~ ADM_RATE + SAT_AVG + net_price + inst_type +
## UGDS + REGION
## Model 2: MD_EARN_WNE_P10 ~ ADM_RATE + SAT_AVG + net_price + inst_type +
```

```
##      UGDS + REGION + stem_share + is_stem_heavy
## Res.Df      RSS Df Sum of Sq      F      Pr(>F)
## 1      974 1.0335e+11
## 2      972 8.8231e+10  2 1.5118e+10 83.276 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Interpretation: If adding field-of-study variables (STEM share) substantially increases R^2 and the partial F-test is significant, H3 is supported — field matters beyond institution alone.

5.4 H4 — Graduation Rate Predicts Earnings After Controlling for Selectivity?

```
h4_model <- lm(MD_EARN_WNE_P10 ~ ADM_RATE + SAT_AVG + C150_4, data = model_df)
summary(h4_model)
```

```
##
## Call:
## lm(formula = MD_EARN_WNE_P10 ~ ADM_RATE + SAT_AVG + C150_4, data = model_df)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -36674  -6321  -1140    4491   80756
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -11457.546   4579.876  -2.502   0.0125 *
## ADM_RATE     -3447.087   1876.122  -1.837   0.0665 .
## SAT_AVG        50.387     4.219   11.942 < 2e-16 ***
## C150_4       22468.061   3432.262   6.546 9.48e-11 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 10800 on 986 degrees of freedom
## Multiple R-squared:  0.5221, Adjusted R-squared:  0.5207
## F-statistic: 359.1 on 3 and 986 DF,  p-value: < 2.2e-16
```

Interpretation: A significant positive coefficient on C150_4 ($p < 0.05$) supports H4 — graduation rate predicts earnings even after controlling for institutional selectivity.

6. Linear Regression Modeling

6.1 Train/Test Split

```
set.seed(42)
train_idx <- createDataPartition(model_df$MD_EARN_WNE_P10, p = 0.8, list = FALSE)
train_df  <- model_df[train_idx, ]
test_df   <- model_df[-train_idx, ]

cat("Training set size:", nrow(train_df), "\n")

## Training set size: 794
```

```
cat("Test set size:    ", nrow(test_df), "\n")
```

```
## Test set size:      196
```

6.2 Multiple Linear Regression

```
lm_full <- lm(
  MD_EARN_WNE_P10 ~ net_price + ADM_RATE + SAT_AVG +
    inst_type + C150_4 + log_ugds + REGION + stem_share,
  data = train_df
)
```

```
summary(lm_full)
```

```
##
## Call:
## lm(formula = MD_EARN_WNE_P10 ~ net_price + ADM_RATE + SAT_AVG +
##     inst_type + C150_4 + log_ugds + REGION + stem_share, data = train_df)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -30586   -4861    -576     3480    82903
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    2.092e+04  1.167e+04   1.792 0.073462 .
## net_price      1.242e-01  5.548e-02   2.238 0.025480 *
## ADM_RATE      -5.591e+03  1.898e+03  -2.946 0.003321 **
## SAT_AVG        1.855e+01  4.628e+00   4.008 6.71e-05 ***
## inst_typePrivate Nonprofit  9.617e+03  5.585e+03   1.722 0.085469 .
## inst_typePublic    6.524e+03  5.655e+03   1.154 0.248942
## C150_4          1.739e+04  3.640e+03   4.778 2.12e-06 ***
## log_ugds        1.319e+03  4.227e+02   3.120 0.001878 **
## REGION1       -1.201e+04  9.553e+03  -1.257 0.208982
## REGION2       -1.376e+04  9.534e+03  -1.443 0.149453
## REGION3       -1.924e+04  9.524e+03  -2.020 0.043705 *
## REGION4       -1.752e+04  9.541e+03  -1.836 0.066734 .
## REGION5       -2.273e+04  9.519e+03  -2.388 0.017195 *
## REGION6       -1.932e+04  9.585e+03  -2.016 0.044136 *
## REGION7       -1.677e+04  9.765e+03  -1.718 0.086282 .
## REGION8       -1.452e+04  9.629e+03  -1.508 0.131913
## REGION9       -4.031e+04  1.056e+04  -3.818 0.000145 ***
## stem_share      3.225e+04  2.601e+03  12.396 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 9333 on 776 degrees of freedom
## Multiple R-squared:  0.6522, Adjusted R-squared:  0.6446
## F-statistic: 85.59 on 17 and 776 DF,  p-value: < 2.2e-16
```

6.3 Forward Stepwise Selection

```

# Fit null and full models for stepwise bounds
lm_null <- lm(MD_EARN_WNE_P10 ~ 1, data = train_df)

lm_step <- stepAIC(lm_null,
                  scope = list(lower = lm_null, upper = lm_full),
                  direction = "forward",
                  trace = FALSE)

cat("Selected predictors:\n")

## Selected predictors:
print(names(coef(lm_step))[-1])

## [1] "SAT_AVG"          "stem_share"
## [3] "REGION1"          "REGION2"
## [5] "REGION3"          "REGION4"
## [7] "REGION5"          "REGION6"
## [9] "REGION7"          "REGION8"
## [11] "REGION9"          "C150_4"
## [13] "net_price"        "ADM_RATE"
## [15] "log_ugds"         "inst_typePrivate Nonprofit"
## [17] "inst_typePublic"

```

6.4 Test Set Evaluation

```

# Predictions on held-out test set
preds_lm <- predict(lm_step, newdata = test_df)

rmse_lm <- sqrt(mean((test_df$MD_EARN_WNE_P10 - preds_lm)^2, na.rm = TRUE))
r2_lm <- cor(test_df$MD_EARN_WNE_P10, preds_lm, use = "complete.obs")^2

cat(sprintf("Test RMSE: $%s\n", format(round(rmse_lm), big.mark = ",")))

## Test RMSE: $9,449

cat(sprintf("Test R²:   %.4f\n", r2_lm))

## Test R²:   0.6246

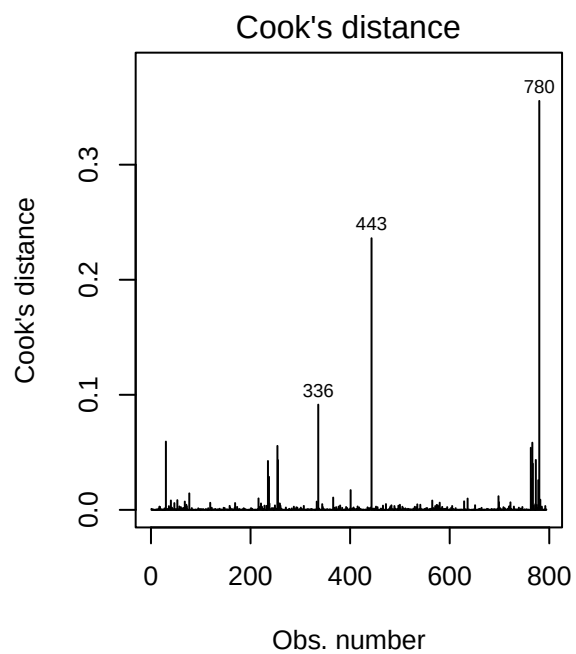
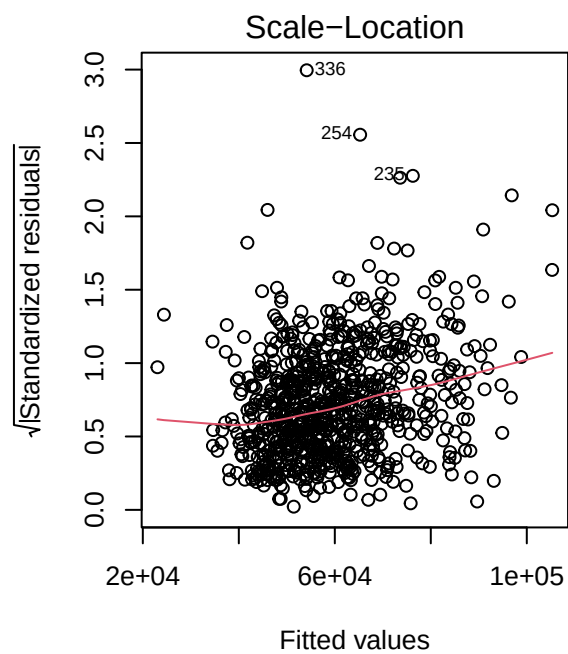
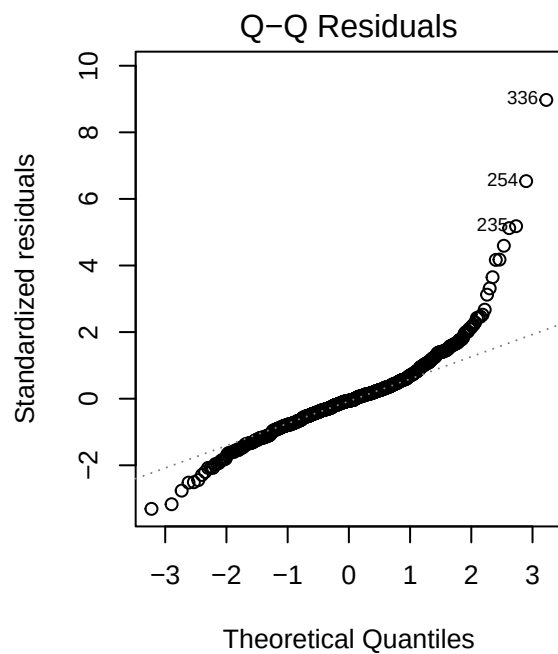
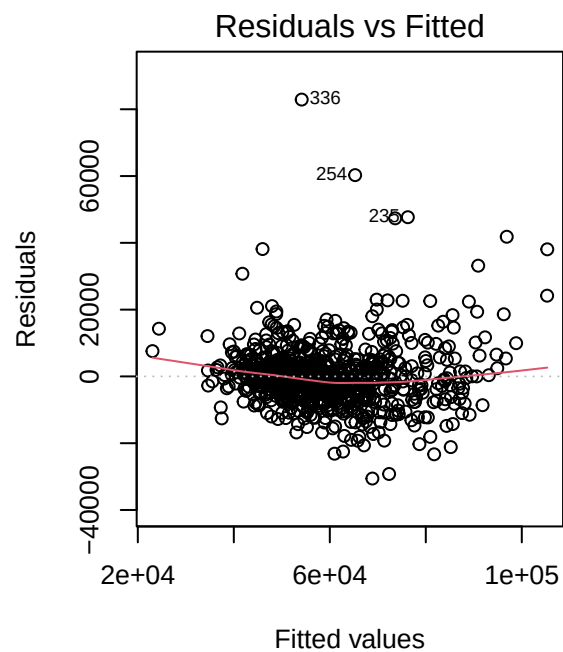
```

6.5 Residual Diagnostics

```

par(mfrow = c(2, 2))
plot(lm_step, which = 1:4)

```



```
par(mfrow = c(1, 1))
```

6.6 K-Fold Cross-Validation

```
ctrl_cv <- trainControl(method = "cv", number = 10, verboseIter = FALSE)

cv_model <- train(
  MD_EARN_WNE_P10 ~ net_price + ADM_RATE + SAT_AVG +
    inst_type + C150_4 + log_ugds + REGION + stem_share,
  data      = model_df,
  method    = "lm",
  trControl = ctrl_cv,
  na.action = na.omit
)

cat("10-Fold CV Results:\n")

## 10-Fold CV Results:
print(cv_model$results[, c("RMSE", "Rsquared", "MAE")])

##           RMSE Rsquared      MAE
## 1  9435.546  0.6338123 6429.999
```

7. ROI Classification (Logistic Regression)

7.1 Class Balance

```
table(model_df$high_roi) |>
  as.data.frame() |>
  setNames(c("ROI Class", "Count")) |>
  mutate(`ROI Class` = ifelse(`ROI Class` == 1, "High ROI", "Low ROI")) |>
  kable(caption = "Distribution of Binary ROI Classes") |>
  kable_styling(bootstrap_options = c("striped", "hover"), full_width = FALSE)
```

Table 3: Distribution of Binary ROI Classes

ROI Class	Count
Low ROI	442
High ROI	548

7.2 Logistic Regression Model

```
train_cls <- train_df |> filter(!is.na(high_roi))
test_cls  <- test_df  |> filter(!is.na(high_roi))

logit_model <- glm(
  high_roi ~ net_price + ADM_RATE + SAT_AVG +
    inst_type + C150_4 + log_ugds + REGION + stem_share,
  data      = train_cls,
  family    = binomial
)
```

```
summary(logit_model)
```

```
##
## Call:
## glm(formula = high_roi ~ net_price + ADM_RATE + SAT_AVG + inst_type +
##      C150_4 + log_ugds + REGION + stem_share, family = binomial,
##      data = train_cls)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)      5.061e+00  8.827e+02   0.006  0.99543
## net_price        -5.644e-04  4.936e-05 -11.434 < 2e-16 ***
## ADM_RATE         -2.446e-01  7.377e-01  -0.332  0.74017
## SAT_AVG           4.775e-03  1.714e-03   2.786  0.00534 **
## inst_typePrivate Nonprofit -1.505e+00  1.635e+00  -0.921  0.35726
## inst_typePublic   -1.989e+00  1.635e+00  -1.216  0.22394
## C150_4            2.357e+00  1.247e+00   1.890  0.05870 .
## log_ugds          4.631e-01  1.696e-01   2.730  0.00633 **
## REGION1          -2.104e+00  8.827e+02  -0.002  0.99810
## REGION2          -1.997e+00  8.827e+02  -0.002  0.99820
## REGION3          -3.210e+00  8.827e+02  -0.004  0.99710
## REGION4          -2.623e+00  8.827e+02  -0.003  0.99763
## REGION5          -3.435e+00  8.827e+02  -0.004  0.99689
## REGION6          -3.457e+00  8.827e+02  -0.004  0.99688
## REGION7          -2.284e+00  8.827e+02  -0.003  0.99794
## REGION8          -1.746e+00  8.827e+02  -0.002  0.99842
## REGION9          -8.993e+00  8.827e+02  -0.010  0.99187
## stem_share        7.265e+00  1.293e+00   5.619 1.92e-08 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 1092.64  on 793  degrees of freedom
## Residual deviance:  426.05  on 776  degrees of freedom
## AIC: 462.05
##
## Number of Fisher Scoring iterations: 13
```

7.3 Predictions & Confusion Matrix

```
probs_logit <- predict(logit_model, newdata = test_cls, type = "response")
preds_logit <- ifelse(probs_logit > 0.5, 1, 0)

cm <- confusionMatrix(
  factor(preds_logit, levels = c(0, 1)),
  factor(test_cls$high_roi, levels = c(0, 1)),
  positive = "1"
)
print(cm)

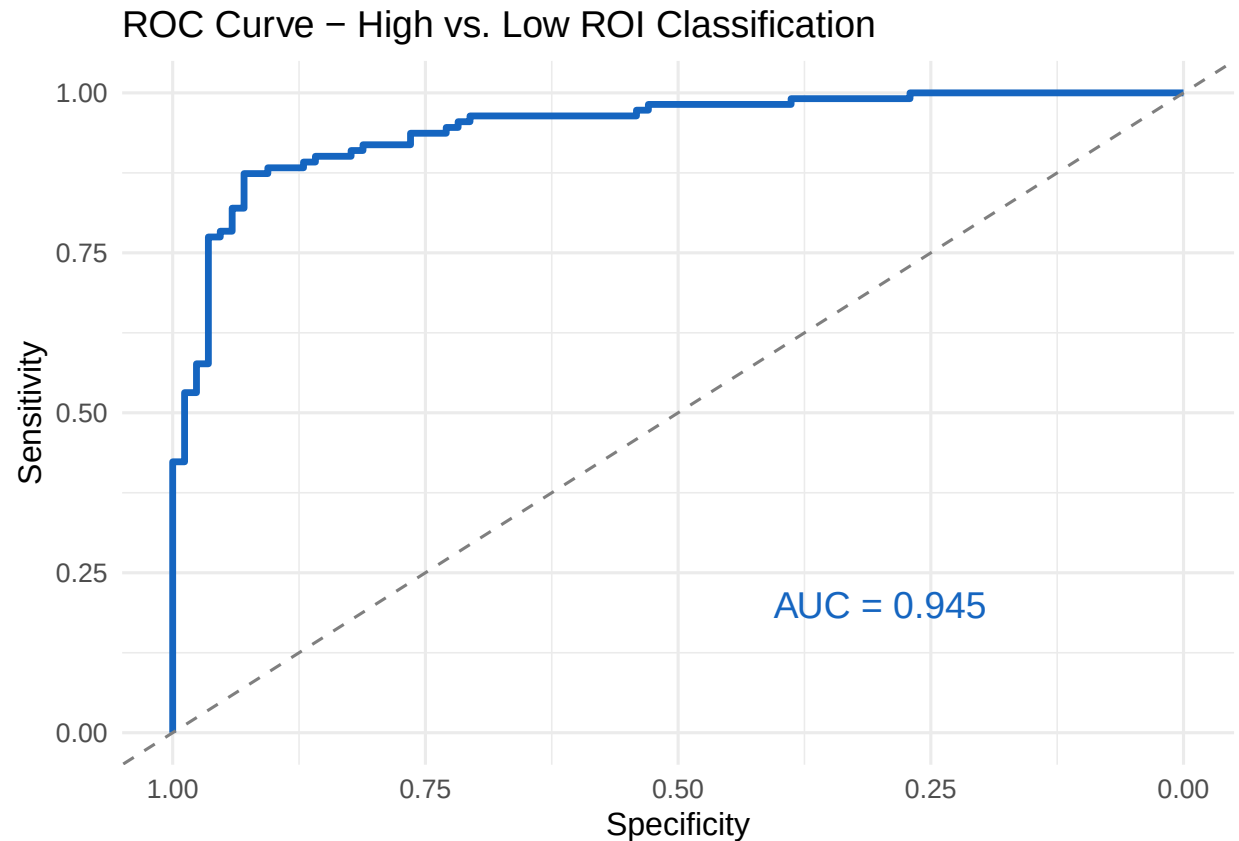
## Confusion Matrix and Statistics
##
```

```
##           Reference
## Prediction    0    1
##           0  70  10
##           1  15 101
##
##           Accuracy : 0.8724
##           95% CI : (0.8175, 0.9157)
##           No Information Rate : 0.5663
##           P-Value [Acc > NIR] : <2e-16
##
##           Kappa : 0.7385
##
## Mcnemar's Test P-Value : 0.4237
##
##           Sensitivity : 0.9099
##           Specificity : 0.8235
##           Pos Pred Value : 0.8707
##           Neg Pred Value : 0.8750
##           Prevalence : 0.5663
##           Detection Rate : 0.5153
##           Detection Prevalence : 0.5918
##           Balanced Accuracy : 0.8667
##
##           'Positive' Class : 1
##
```

7.4 ROC Curve & AUC

```
roc_obj <- roc(test_cls$high_roi, probs_logit, quiet = TRUE)

ggroc(roc_obj, colour = "#1565C0", linewidth = 1.2) +
  geom_abline(slope = 1, intercept = 1, linetype = "dashed", color = "gray50") +
  annotate("text", x = 0.3, y = 0.2,
    label = sprintf("AUC = %.3f", auc(roc_obj)),
    size = 5, color = "#1565C0") +
  labs(title = "ROC Curve - High vs. Low ROI Classification",
    x = "Specificity", y = "Sensitivity") +
  theme_minimal(base_size = 12)
```



```
cat(sprintf("AUC: %.4f\n", auc(roc_obj)))
```

```
## AUC: 0.9454
```

8. Dimension Reduction & Clustering

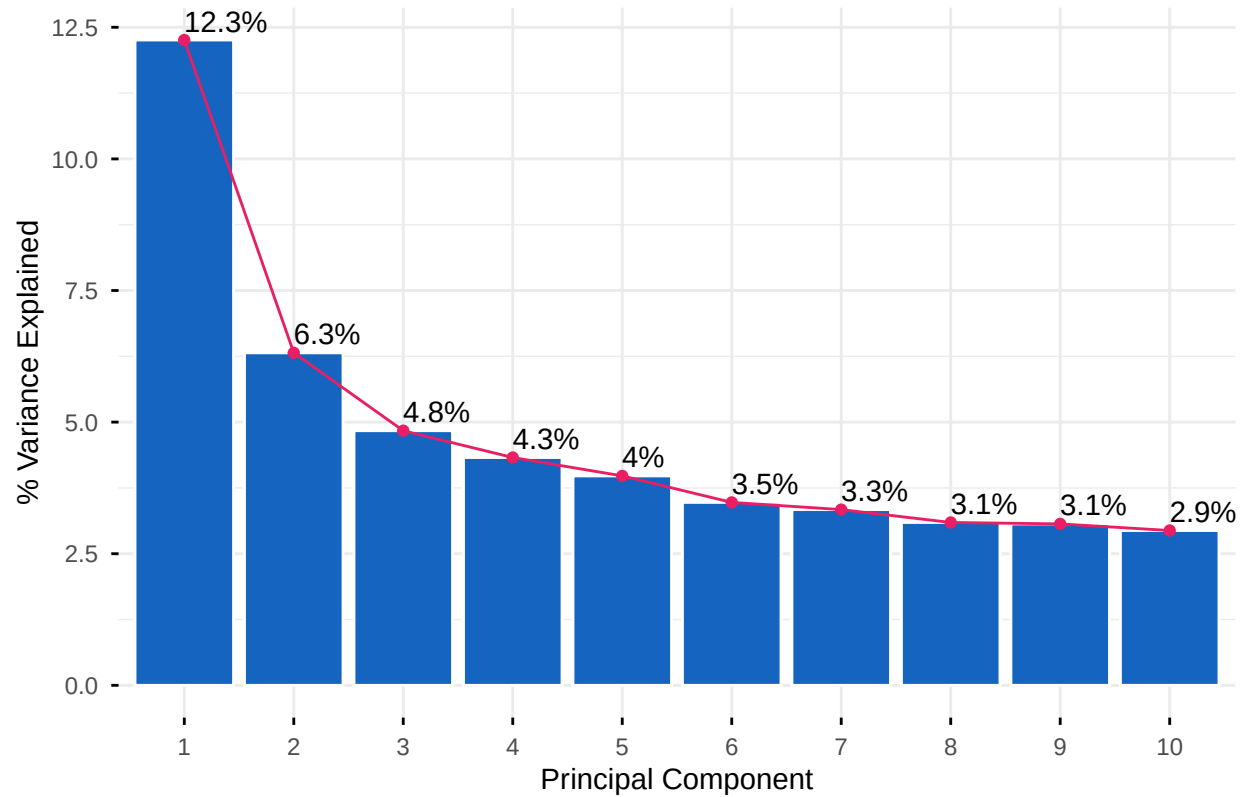
8.1 PCA on Field-Share Variables

```
pcip_data <- model_df |>
  select(starts_with("PCIP")) |>
  select(where(~mean(is.na(.)) < 0.4)) |>
  na.omit()

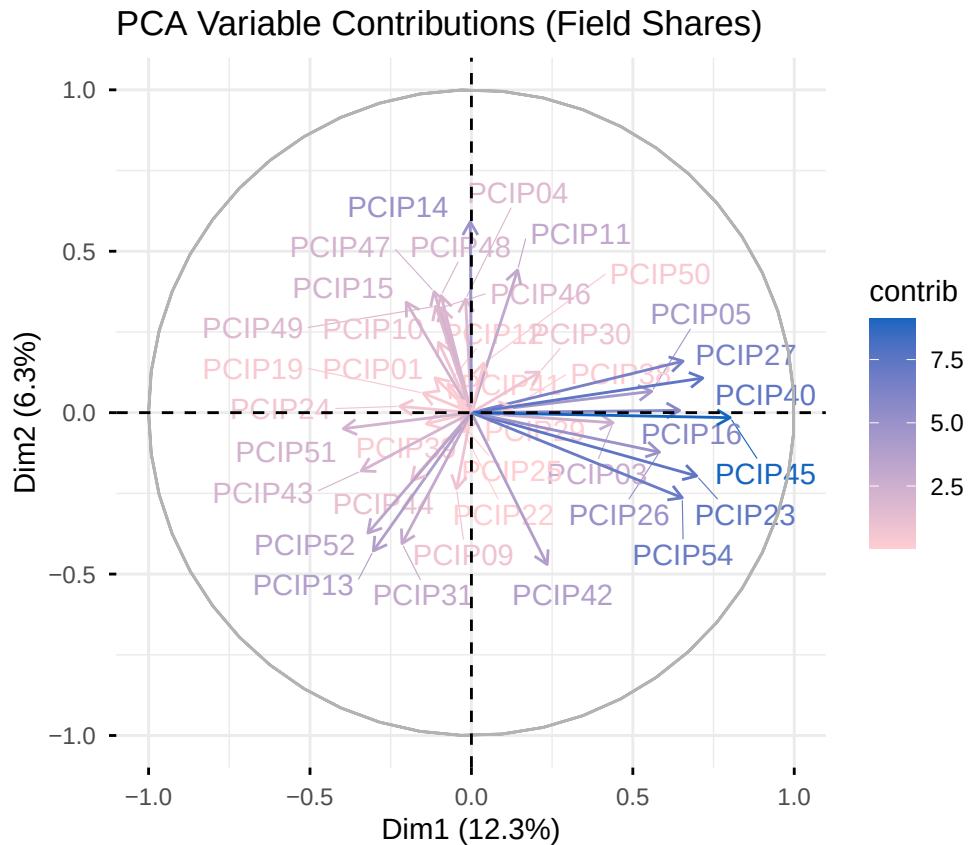
pca_result <- prcomp(pcip_data, center = TRUE, scale. = TRUE)

# Variance explained
fviz_eig(pca_result,
  addlabels = TRUE,
  barfill = "#1565C0",
  barcolor = "white",
  linecolor = "#E91E63") +
  labs(title = "PCA Scree Plot - Field-Share Variables",
    x = "Principal Component", y = "% Variance Explained")
```

PCA Scree Plot – Field–Share Variables



```
fviz_pca_var(pca_result,
  col.var = "contrib",
  gradient.cols = c("#FFCDD2", "#1565C0"),
  repel = TRUE,
  title = "PCA Variable Contributions (Field Shares)")
```



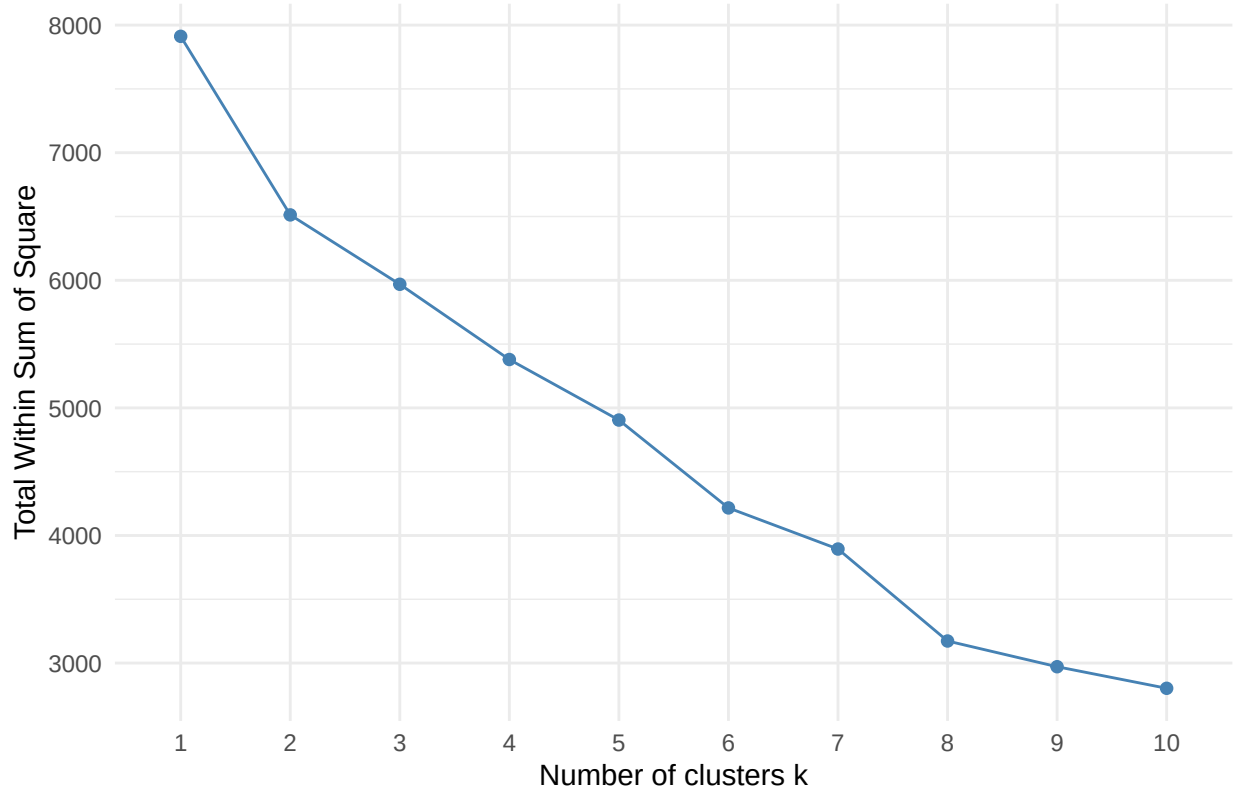
8.2 K-Means Clustering

```
# Build clustering feature matrix: PCA scores + cost/selectivity/outcome
pca_scores <- as.data.frame(pca_result$x[, 1:5])
inst_for_cluster <- model_df[complete.cases(model_df[, c("net_price", "SAT_AVG", "ROI")]), ]
n_cluster_rows <- min(nrow(pca_scores), nrow(inst_for_cluster))

cluster_features <- bind_cols(
  pca_scores[1:n_cluster_rows, ],
  inst_for_cluster[1:n_cluster_rows, c("net_price", "SAT_AVG", "ROI")]
) |> scale()

# Elbow method
fviz_nbclust(cluster_features, kmeans, method = "wss", k.max = 10) +
  labs(title = "Elbow Method - Optimal Number of Clusters") +
  theme_minimal()
```

Elbow Method – Optimal Number of Clusters



```
set.seed(42)
km_fit <- kmeans(cluster_features, centers = 4, nstart = 25, iter.max = 100)

# Attach cluster labels to institutions
clustered_df <- inst_for_cluster[1:n_cluster_rows, ] |>
  mutate(cluster = factor(km_fit$cluster))

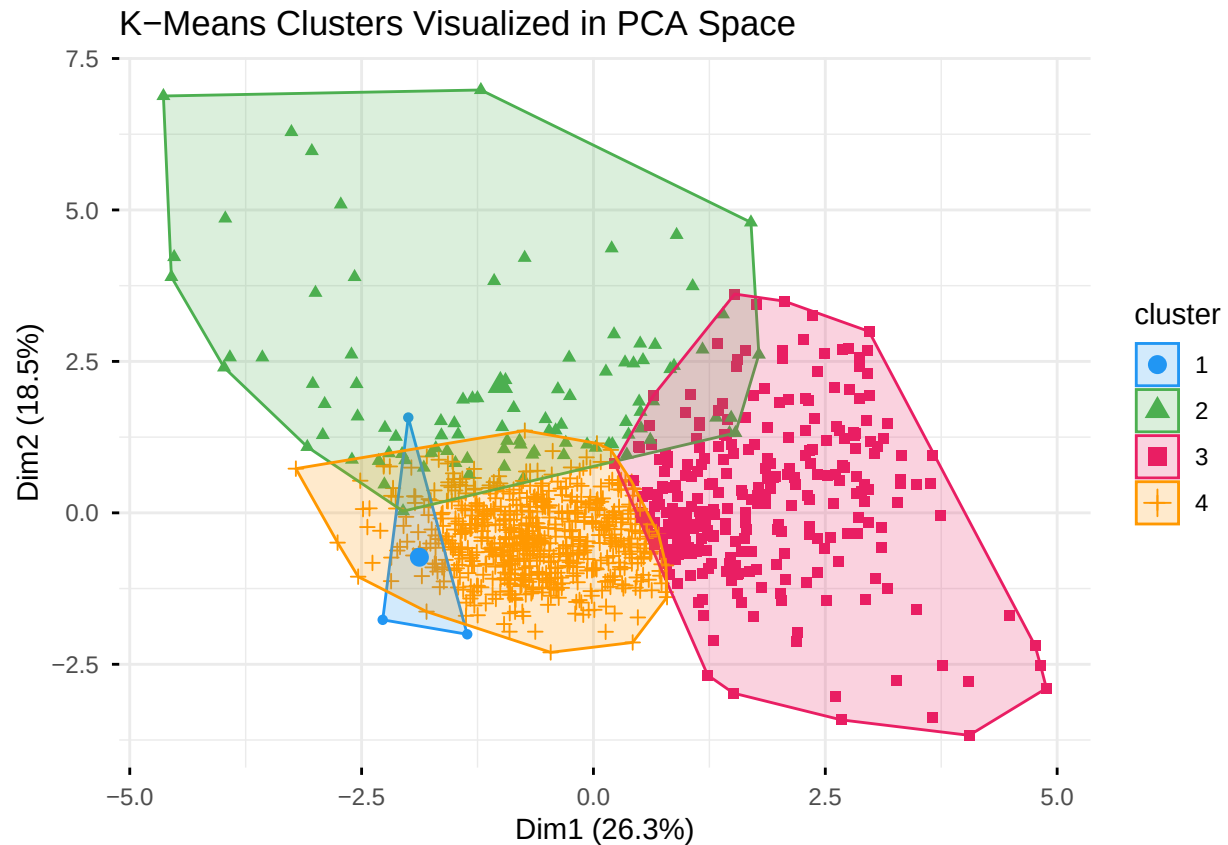
cat("Cluster sizes:\n")

## Cluster sizes:
print(table(clustered_df$cluster))
```

```
##
##  1  2  3  4
##  3 97 279 611
```

8.3 Cluster Visualization in PCA Space

```
fviz_cluster(km_fit,
  data = cluster_features,
  geom = "point",
  ellipse.type = "convex",
  palette = c("#2196F3", "#4CAF50", "#E91E63", "#FF9800"),
  ggtheme = theme_minimal() +
  labs(title = "K-Means Clusters Visualized in PCA Space")
```



8.4 Cluster Profiles

```
cluster_profiles <- clustered_df |>
  group_by(cluster) |>
  summarise(
    n = n(),
    median_roi = round(median(ROI, na.rm = TRUE), 2),
    median_earn = round(median(MD_EARN_WNE_P10, na.rm = TRUE)),
    median_cost = round(median(net_price, na.rm = TRUE)),
    median_sat = round(median(SAT_AVG, na.rm = TRUE)),
    pct_public = round(mean(CONTROL == 1, na.rm = TRUE) * 100, 1),
    pct_forprofit = round(mean(CONTROL == 3, na.rm = TRUE) * 100, 1),
    .groups = "drop"
  )

cluster_profiles |>
  kable(
    col.names = c("Cluster", "N", "Median ROI", "Median Earnings ($)",
                  "Median Net Price ($)", "Median SAT", "% Public", "% For-Profit"),
    caption = "Cluster Profiles: Cost, Selectivity, and Outcomes",
    format.args = list(big.mark = ",")
  ) |>
  kable_styling(bootstrap_options = c("striped", "hover"))
```

Table 4: Cluster Profiles: Cost, Selectivity, and Outcomes

Cluster	N	Median ROI	Median Earnings (\$)	Median Net Price (\$)	Median SAT	% Public	% For-Profit
1	3	3.03	51,629	17,225	1,055	66.7	0.0
2	97	5.81	64,010	10,974	1,200	81.4	1.0
3	279	2.29	69,498	29,519	1,335	12.2	0.7
4	611	2.87	52,581	18,600	1,114	47.3	0.3

```
# Top schools by ROI in each cluster
clustered_df |>
  group_by(cluster) |>
  slice_max(ROI, n = 3) |>
  select(cluster, INSTNM, inst_type, ROI, MD_EARN_WNE_P10, net_price) |>
  mutate(across(c(ROI), round, 2),
         across(c(MD_EARN_WNE_P10, net_price), ~dollar(round(.)))) |>
  kable(col.names = c("Cluster", "Institution", "Type", "ROI",
                     "Earnings (10yr)", "Net Price"),
        caption = "Top 3 Schools by ROI Within Each Cluster") |>
  kable_styling(bootstrap_options = c("striped", "hover"))
```

Table 5: Top 3 Schools by ROI Within Each Cluster

Cluster	Institution	Type	ROI	Earnings (10yr)	Net Price
1	Embry-Riddle Aeronautical University-Worldwide	Private Nonprofit	4.49	\$84,131	\$18,725
1	Colorado Mesa University	Public	3.03	\$45,823	\$15,103
1	SUNY College of Technology at Delhi	Public	3.00	\$51,629	\$17,225
2	CUNY Bernard M Baruch College	Public	25.05	\$75,971	\$3,033
2	CUNY Hunter College	Public	21.17	\$63,163	\$2,984
2	CUNY Brooklyn College	Public	19.58	\$60,752	\$3,103
3	University of Chicago	Private Nonprofit	6.18	\$91,885	\$14,860
3	Vanderbilt University	Private Nonprofit	5.78	\$91,565	\$15,846
3	Bowdoin College	Private Nonprofit	5.75	\$82,735	\$14,398
4	University of Akron Wayne College	Public	7.73	\$46,600	\$6,032
4	Baptist Health Sciences University	Private Nonprofit	6.47	\$72,529	\$11,212
4	University of Science and Arts of Oklahoma	Public	6.33	\$41,913	\$6,624

9. Discussion & Conclusions

9.1 Hypothesis Summary

```
tribble(
  ~Hypothesis, ~Prediction, ~Result,
  "H1", "Net price predicts earnings after controlling for selectivity",
  "See Section 5.1 - coefficient on net_price in full model",
  "H2", "For-profit schools deliver lower ROI than public/nonprofit",
  "See Section 5.2 - ANOVA and Tukey post-hoc results",
  "H3", "Field of study is a stronger predictor than institution characteristics",
  "See Section 5.3 - R2 comparison between models",
```

```

"H4", "Higher graduation rates → better earnings after controlling for selectivity",
"See Section 5.4 - coefficient on C150_4"
) |>
kable(caption = "Hypothesis Testing Summary") |>
kable_styling(bootstrap_options = c("striped", "hover"))

```

Table 6: Hypothesis Testing Summary

Hypothesis	Prediction	Result
H1	Net price predicts earnings after controlling for selectivity	See Section 5.1 — coefficient on
H2	For-profit schools deliver lower ROI than public/nonprofit	See Section 5.2 — ANOVA and
H3	Field of study is a stronger predictor than institution characteristics	See Section 5.3 — R^2 comparison
H4	Higher graduation rates → better earnings after controlling for selectivity	See Section 5.4 — coefficient on

9.2 Key Findings

- **ROI varies substantially** by institution type, with for-profit schools generally clustering at lower earnings relative to cost (consistent with preliminary EDA).
- **Net price is not always a proxy for quality** — after controlling for selectivity, additional cost does not uniformly translate to higher earnings.
- **Field of study drives significant variation** in outcomes, supporting the view that what you study matters alongside where you study.
- **Graduation rates signal institutional quality** beyond selectivity alone, suggesting that student support structures have measurable economic value.

9.3 Limitations

- Data is at the institution level; individual-level variation within schools cannot be captured.
- Earnings at 10 years may reflect cohort-specific economic conditions, not just school quality.
- Suppressed values for small programs introduce systematic missingness.
- Causality cannot be inferred — selection effects remain a fundamental challenge.

9.4 Policy Implications

These findings can help prospective students compare programs on a cost-adjusted basis, and encourage policymakers to consider ROI metrics (not just prestige) when designing federal aid policy.

10. References

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